



Telecom Academy



Module 8: Network Layer

Telecom Academy
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Introduction to Networks v7.0
(ITN)



Module 8: Topics

What will I learn to do in this module?

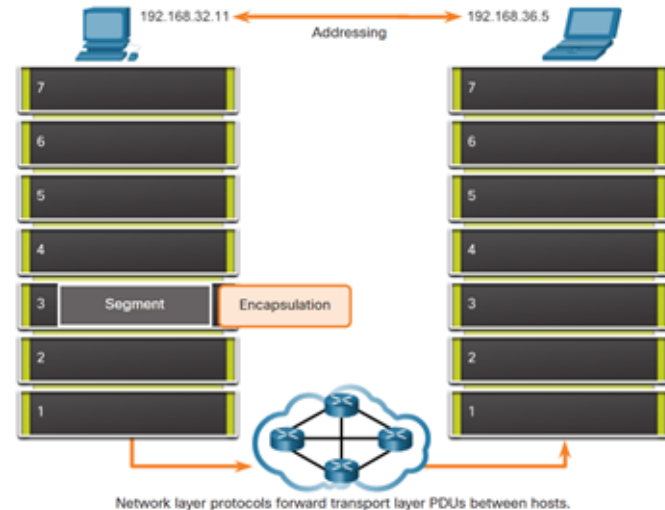
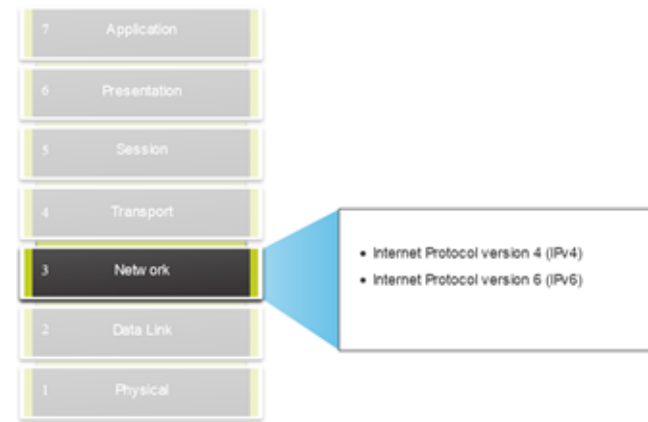
Topic Title	Topic Objective
Network Layer Characteristics	Explain how the network layer uses IP protocols for reliable communications.
IPv4 Packet	Explain the role of the major header fields in the IPv4 packet.
IPv6 Packet	Explain the role of the major header fields in the IPv6 packet.
How a Host Routes	Explain how network devices use routing tables to direct packets to a destination network.
Router Routing Tables	Explain the function of fields in the routing table of a router.

8.1 Network Layer Characteristics

Network Layer Characteristics

The Network Layer

- Provides services to allow end devices to exchange data
- IP version 4 (IPv4) and IP version 6 (IPv6) are the principle network layer communication protocols.
- The network layer performs four basic operations:
 - Addressing end devices
 - Encapsulation
 - Routing
 - De-encapsulation



IP Encapsulation

- IP encapsulates the transport layer segment.
- IP can use either an IPv4 or IPv6 packet and not impact the layer 4 segment.
- IP packet will be examined by all layer 3 devices as it traverses the network.
- The IP addressing does not change from source to destination.

Note: NAT will change addressing, but will be discussed in a later module.

Transport Layer Encapsulation



Transport Layer PDU

Network Layer Encapsulation



Network Layer PDU

IP Packet

Network Layer Characteristics

Characteristics of IP

IP is meant to have low overhead and may be described as:

- Connectionless
- Best Effort
- Media Independent

Network Layer Characteristics

Connectionless

IP is Connectionless

- IP does not establish a connection with the destination before sending the packet.
- There is no control information needed (synchronizations, acknowledgments, etc.).
- The destination will receive the packet when it arrives, but no pre-notifications are sent by IP.
- If there is a need for connection-oriented traffic, then another protocol will handle this (typically TCP at the transport layer).



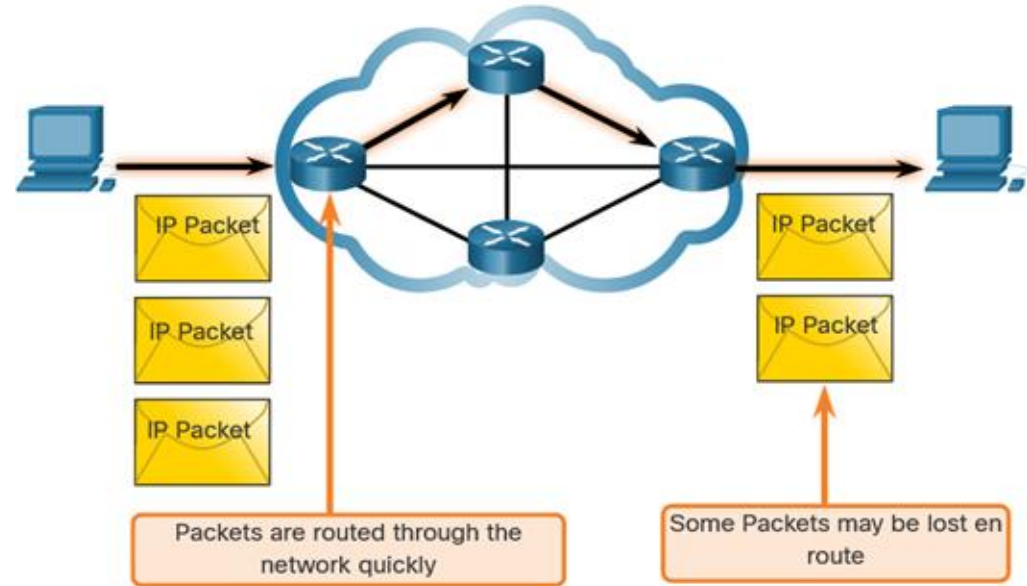
A letter is sent.

Network Layer Characteristics

Best Effort

IP is Best Effort

- IP will not guarantee delivery of the packet.
- IP has reduced overhead since there is no mechanism to resend data that is not received.
- IP does not expect acknowledgments.
- IP does not know if the other device is operational or if it received the packet.



Network Layer Characteristics

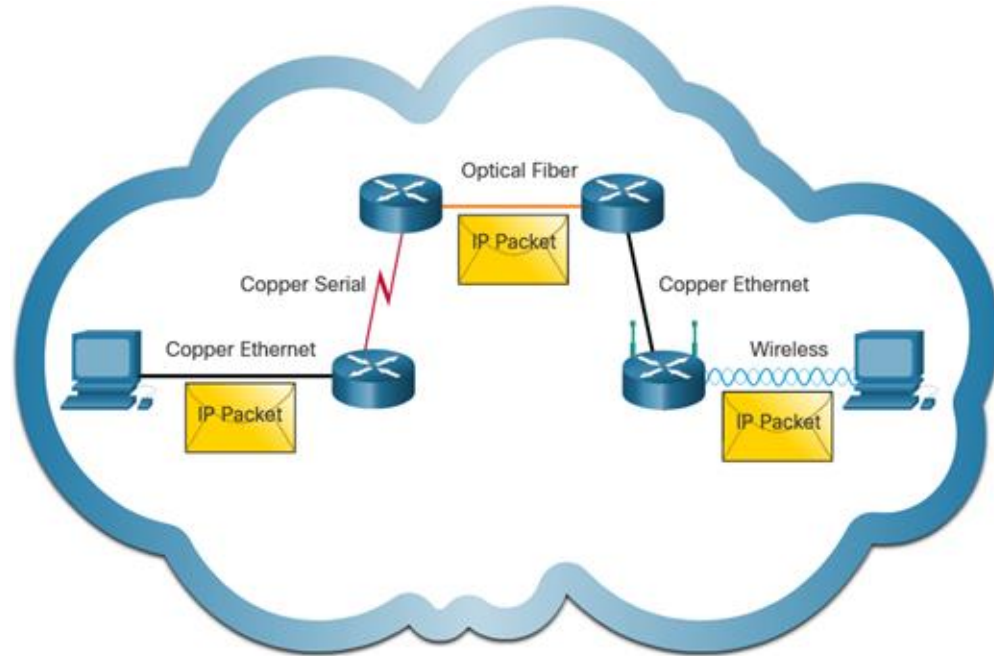
Media Independent

IP is unreliable:

- It cannot manage or fix undelivered or corrupt packets.
- IP cannot retransmit after an error.
- IP cannot realign out of sequence packets.
- IP must rely on other protocols for these functions.

IP is media Independent:

- IP does not concern itself with the type of frame required at the data link layer or the media type at the physical layer.
- IP can be sent over any media type: copper, fiber, or wireless.



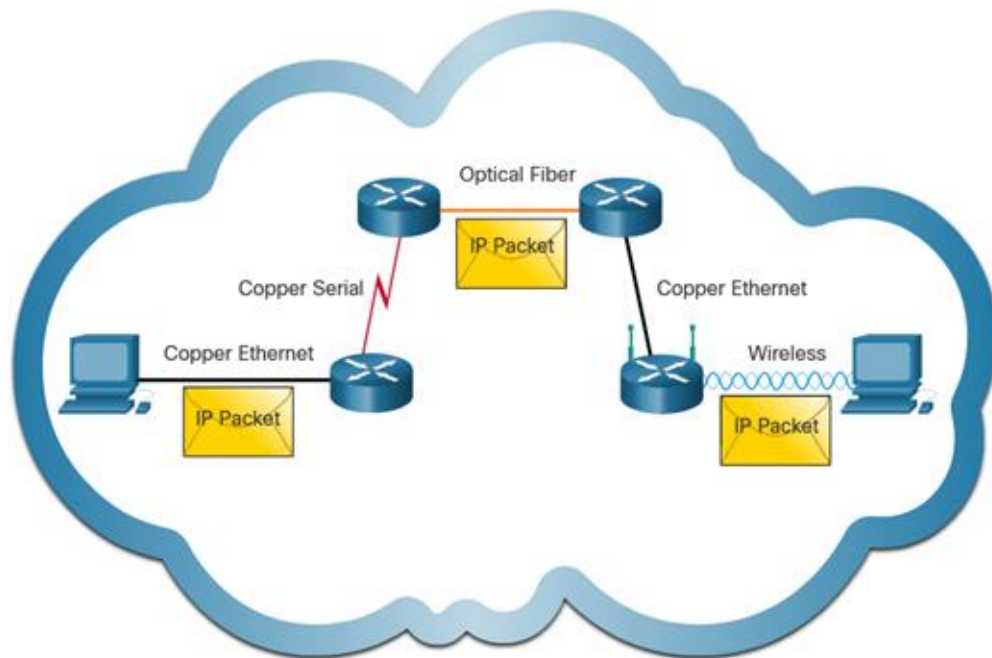
Media Independent (Contd.)

The network layer will establish the Maximum Transmission Unit (MTU).

- Network layer receives this from control information sent by the data link layer.
- The network then establishes the MTU size.

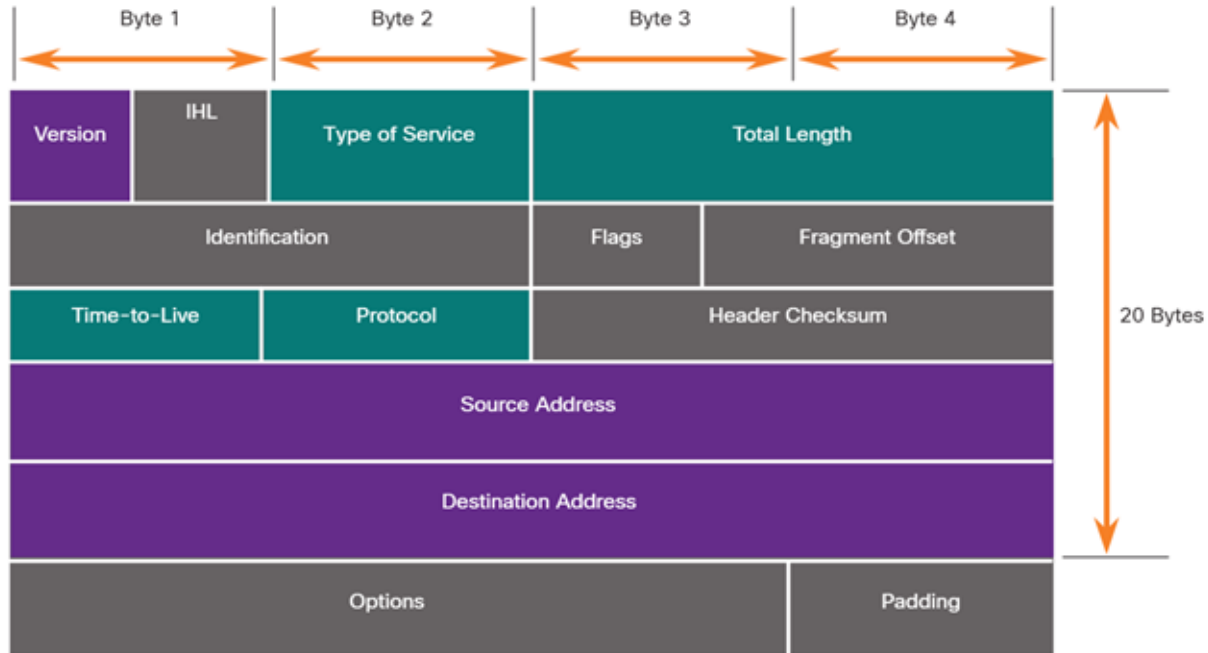
Fragmentation is when Layer 3 splits the IPv4 packet into smaller units.

- Fragmenting causes latency.
- IPv6 does not fragment packets.
- Example: Router goes from Ethernet to a slow WAN with a smaller MTU



8.2 IPv4 Packet

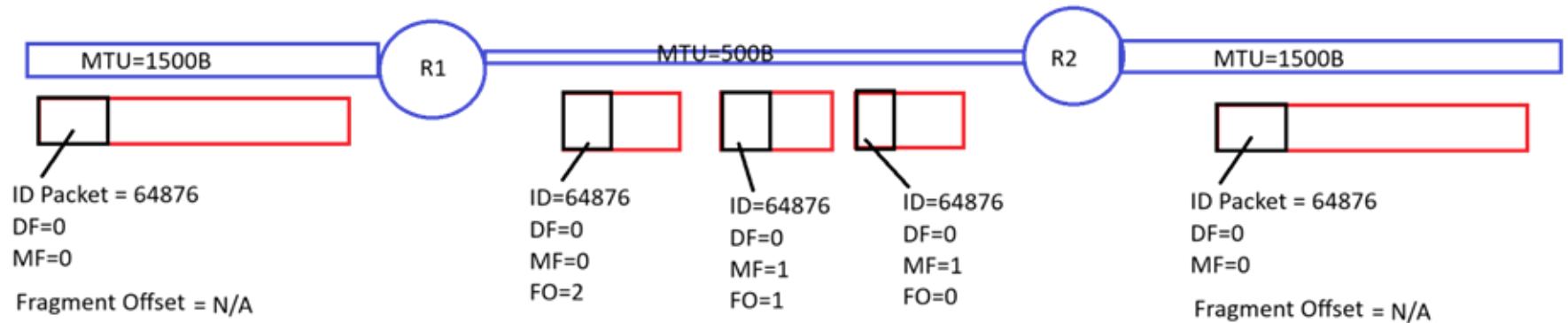
IPv4 Packet Header Fields



Legend

- Fields name kept from IPv4 to IPv6
- Name and position changed in IPv6
- Fields no longer required in IPv6.

IPv4 Packet Fragmentare



8.3 IPv6 Packets

Limitations of IPv4

IPv4 has three major limitations:

- IPv4 address depletion – We have basically run out of IPv4 addressing.
- Lack of end-to-end connectivity – To make IPv4 survive this long, private addressing and NAT were created. This ended direct communications with public addressing.
- Increased network complexity – NAT was meant as temporary solution and creates issues on the network as a side effect of manipulating the network headers addressing. NAT causes latency and troubleshooting issues.

IPv6 Overview

- IPv6 was developed by Internet Engineering Task Force (IETF).
- IPv6 overcomes the limitations of IPv4.
- Improvements that IPv6 provides:
 - **Increased address space** – based on 128 bit address, not 32 bits
 - **Improved packet handling** – simplified header with fewer fields
 - **Eliminates the need for NAT** – since there is a huge amount of addressing, there is no need to use private addressing internally and be mapped to a shared public address

IPv4 and IPv6 Address Space Comparison

Number Name	Scientific Notation	Number of Zeros
1 Thousand	10^3	1,000
1 Million	10^6	1,000,000
1 Billion	10^9	1,000,000,000
1 Trillion	10^{12}	1,000,000,000,000
1 Quadrillion	10^{15}	1,000,000,000,000,000
1 Quintillion	10^{18}	1,000,000,000,000,000,000
1 Sextillion	10^{21}	1,000,000,000,000,000,000,000
1 Septillion	10^{24}	1,000,000,000,000,000,000,000,000
1 Octillion	10^{27}	1,000,000,000,000,000,000,000,000,000
1 Nonillion	10^{30}	1,000,000,000,000,000,000,000,000,000,000
1 Decillion	10^{33}	1,000,000,000,000,000,000,000,000,000,000,000
1 Undecillion	10^{36}	1,000,000,000,000,000,000,000,000,000,000,000,000

Legend



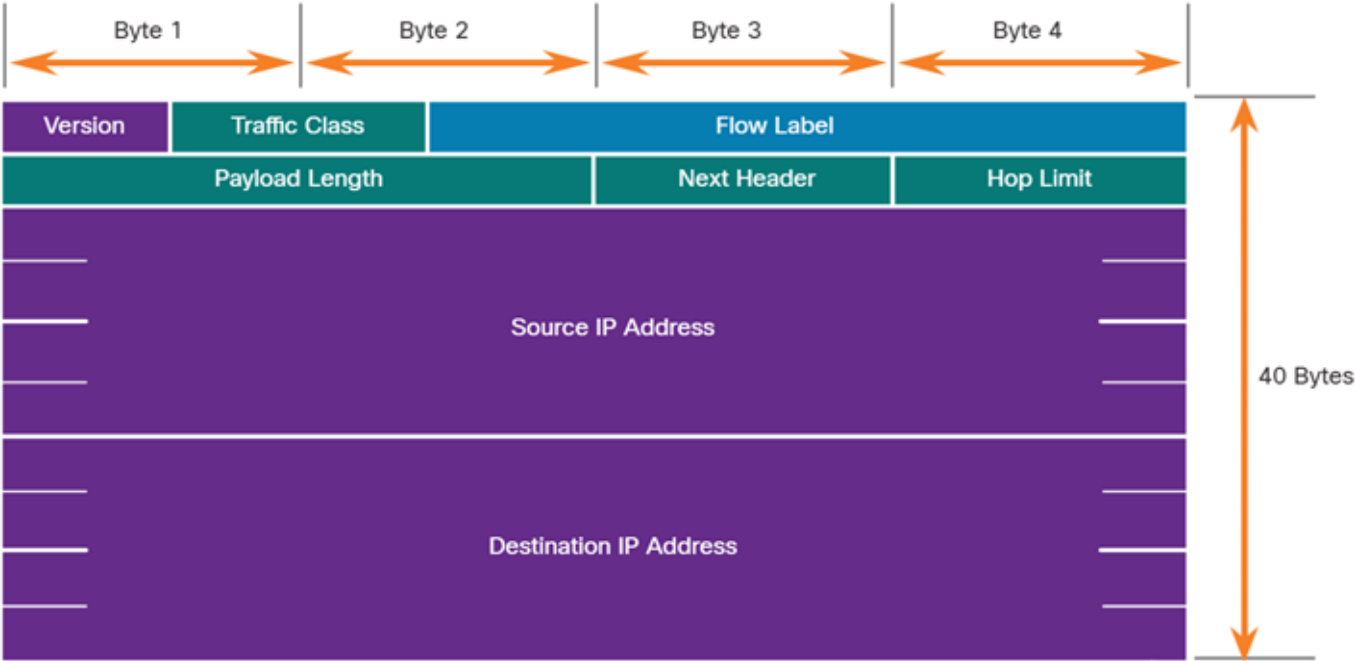
There are 4 billion IPv4 addresses



There are 340 undecillion IPv6 addresses

IPv6 Packets

IPv6 Packet Header



IPv6 Packet Header

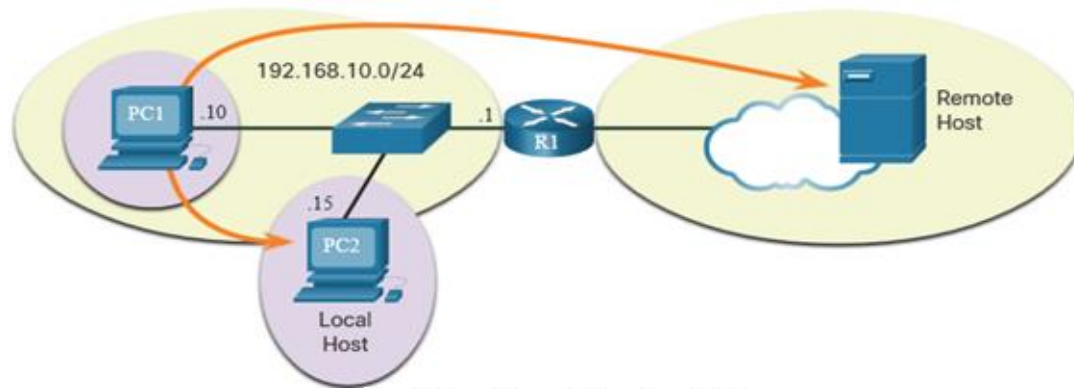
Significant fields in the IPv6 header:

Function	Description
Version	This will be for v6, as opposed to v4, a 4 bit field= 0110
Traffic Class	Used for QoS: Equivalent to DiffServ – DS field
Flow Label	Informs device to handle identical flow labels the same way, 20 bit field
Payload Length	This 16-bit field indicates the length of the data portion or payload of the IPv6 packet
Next Header	I.D.s next level protocol: ICMP, TCP, UDP, etc.
Hop Limit	Replaces TTL field Layer 3 hop count
Source IPv6 Address	128 bit source address
Destination IPV6 Address	128 bit destination address

8.4 How a Host Routes

How a Host Routes

Host Forwarding Decision



IPv4 Routing Table for PC1

```
C:\Users\PC1> netstat -r
```

IPv4 Route Table

Active Routes:

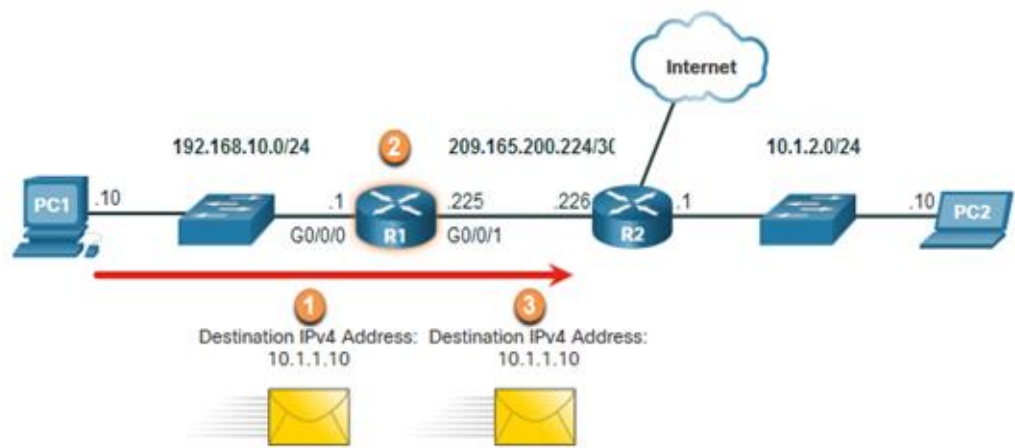
Network	Destination	Netmask	Gateway	Interface	Metric
	0.0.0.0	0.0.0.0	192.168.10.1	192.168.10.10	25
	127.0.0.0	255.0.0.0	On-link	127.0.0.1	306
	127.0.0.1	255.255.255.255	On-link	127.0.0.1	306
127.255.255.255	255.255.255.255	255.255.255.255	On-link	127.0.0.1	306
	192.168.10.0	255.255.255.0	On-link	192.168.10.10	281
	192.168.10.10	255.255.255.255	On-link	192.168.10.10	281
	192.168.10.255	255.255.255.255	On-link	192.168.10.10	281
	224.0.0.0	240.0.0.0	On-link	127.0.0.1	306
	224.0.0.0	240.0.0.0	On-link	192.168.10.10	281
255.255.255.255	255.255.255.255	255.255.255.255	On-link	127.0.0.1	306
255.255.255.255	255.255.255.255	255.255.255.255	On-link	192.168.10.10	281

8.5 Introduction to Routing

Introduction to Routing

Router Packet Forwarding Decision

What happens when the router receives the frame from the host device?



R1 Routing Table

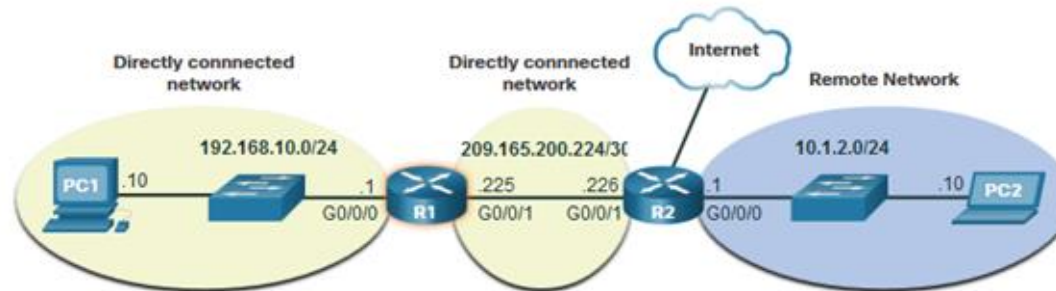
Route	Next Hop or Exit Interface
192.168.10.0 /24	G0/0/0
209.165.200.224/30	G0/0/1
10.1.1.0/24	via R2
Default Route 0.0.0.0/0	via R2

1. Packet arrives on the Gigabit Ethernet 0/0/0 interface of router R1. R1 de-encapsulates the Layer 2 Ethernet header and trailer.
2. Router R1 examines the destination IPv4 address of the packet and searches for the best match in its IPv4 routing table. The route entry indicates that this packet is to be forwarded to router R2.
3. Router R1 encapsulates the packet into a new Ethernet header and trailer, and forwards the packet to the next hop router R2.

IP Router Routing Table

There are three types of routes in a router's routing table:

- **Directly Connected** – These routes are automatically added by the router, provided the interface is active and has addressing.
- **Remote** – These are the routes the router does not have a direct connection and may be learned:
 - Manually – with a static route
 - Dynamically – by using a routing protocol to have the routers share their information with each other
- **Default Route** – this forwards all traffic to a specific direction when there is not a match in the routing table

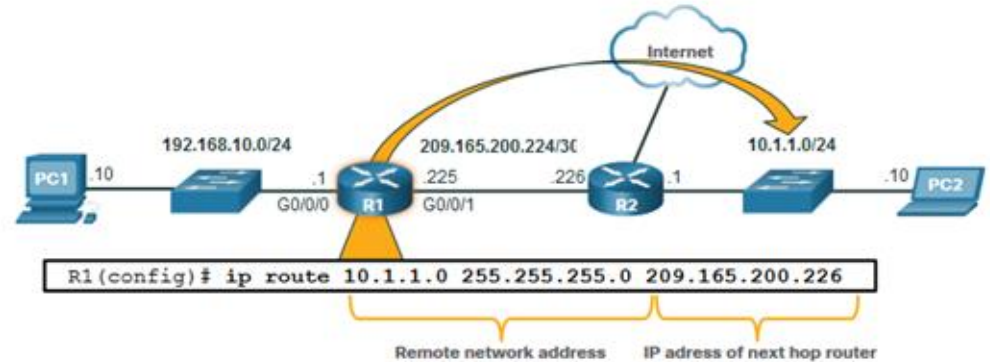


Introduction to Routing

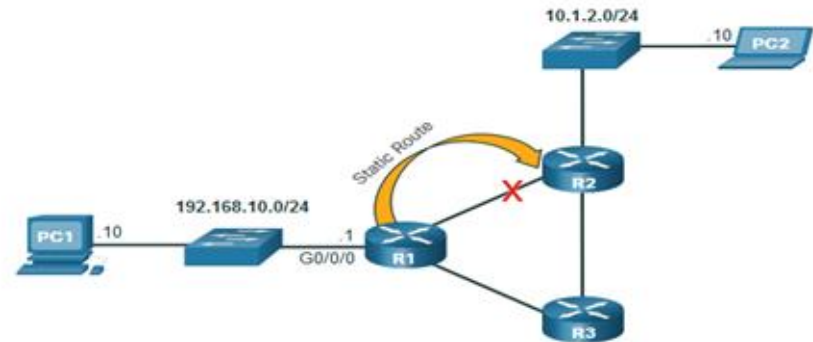
Static Routing

Static Route Characteristics:

- Must be configured manually
- Must be adjusted manually by the administrator when there is a change in the topology
- Good for small non-redundant networks
- Often used in conjunction with a dynamic routing protocol for configuring a default route



R1 is manually configured with a static route to reach the 10.1.1.0/24 network. If this path changes, R1 will require a new static route.



If the route from R1 via R2 is no longer available, a new static route via R3 would need to be configured. A static route does not automatically adjust for topology changes.

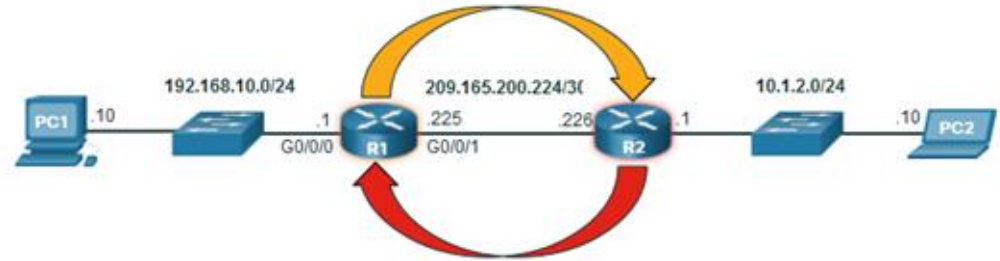
Introduction to Routing

Dynamic Routing

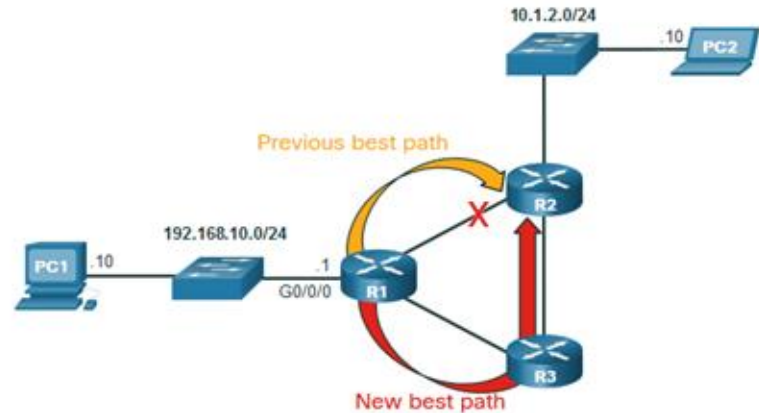
Dynamic Routes Automatically:

- Discover remote networks
- Maintain up-to-date information
- Choose the best path to the destination
- Find new best paths when there is a topology change

Dynamic routing can also share static default routes with the other routers.



- R1 is using the routing protocol OSPF to let R2 know about the 192.168.10.0/24 network.
- R2 is using the routing protocol OSPF to let R1 know about the 10.1.1.0/24 network.

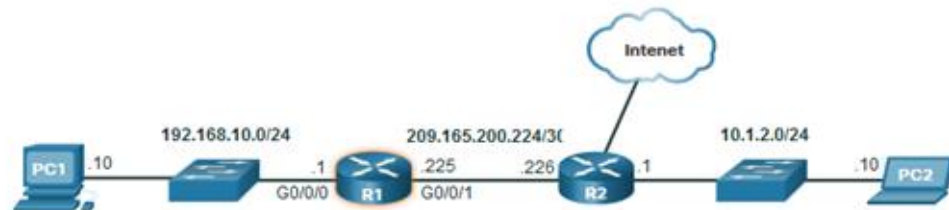


R1, R2, and R3 are using the dynamic routing protocol OSPF. If there is a network topology change, they can automatically adjust to find a new best path.

Introduction to an IPv4 Routing Table

The **show ip route** command shows the following route sources:

- **L** - Directly connected local interface IP address
- **C** – Directly connected network
- **S** – Static route was manually configured by an administrator
- **O** – OSPF
- **D** – EIGRP



```
R1# show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PBR

Gateway of last resort is 209.165.200.226 to network 0.0.0.0
S*   0.0.0.0/0 [1/0] via 209.165.200.226, GigabitEthernet0/0/1
     10.0.0.0/24 is subnetted, 1 subnets
O    10.1.1.0 [110/2] via 209.165.200.226, 00:02:45, GigabitEthernet0/0/1
     192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0/0
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0/0
     209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C    209.165.200.0/24 is directly connected, GigabitEthernet0/0/1
L    209.165.200.225/32 is directly connected, GigabitEthernet0/0/1
R1#
```

This command shows types of routes:

- Directly Connected – C and L
- Remote Routes – O, D, etc.
- Default Routes – S*



Module 13: ICMP

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Introduction to Networks v7.0
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Module Objectives

Module Title: ICMP

Module Objective: Use various tools to test network connectivity.

Topic Title	Topic Objective
ICMP Messages	Explain how ICMP is used to test network connectivity.
Ping and Traceroute Testing	Use ping and traceroute utilities to test network connectivity.

13.1 ICMP Messages

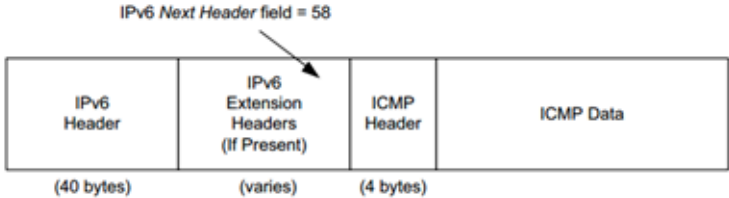
ICMPv4 and ICMPv6 Messages

- Internet Control Message Protocol (ICMP) provides feedback about issues related to the processing of IP packets under certain conditions.
- ICMPv4 is the messaging protocol for IPv4. ICMPv6 is the messaging protocol for IPv6 and includes additional functionality.
- The ICMP messages common to both ICMPv4 and ICMPv6 include:
 - Host reachability
 - Destination or Service Unreachable
 - Time exceeded

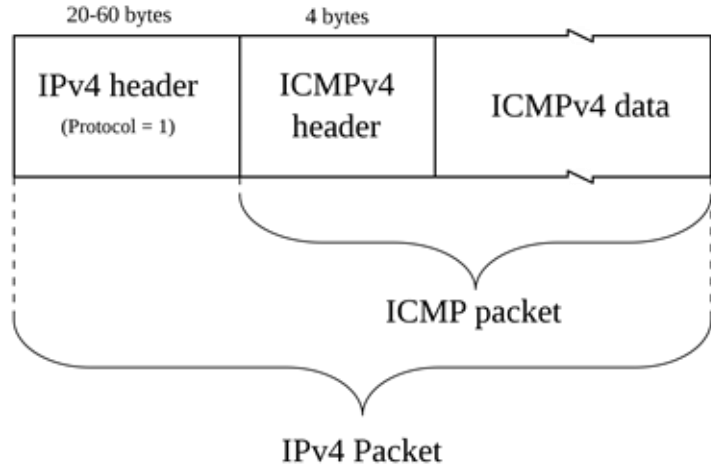
Note: ICMPv4 messages are not required and are often not allowed within a network for security reasons.

ICMP Messages

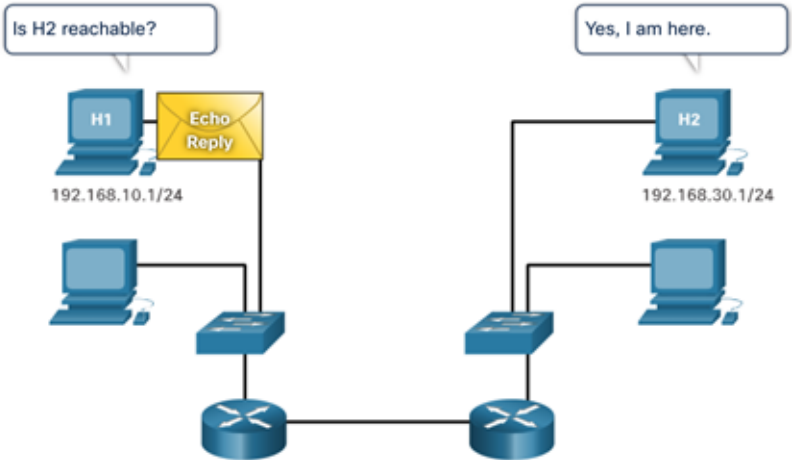
Host Reachability



IPv4



IPv6



type	code	Description	Handled by
0	0	echo reply	user process
3	0	destination unreachable:	
	0	network unreachable	"No route to host"
	1	host unreachable	"No route to host"
	2	protocol unreachable	"Connection refused"
	3	port unreachable	"Connection refused"
	4	fragmentation needed but DF bit set	"Message too long"
	5	source route failed	"No route to host"
	6	destination network unknown	"No route to host"
	7	destination host unknown	"No route to host"
	8	source host isolated (obsolete)	"No route to host"
	9	dest. network administratively prohibited	"No route to host"
	10	dest. host administratively prohibited	"No route to host"
	11	network unreachable for TOS	"No route to host"
	12	host unreachable for TOS	"No route to host"
	13	communication administratively prohibited	(ignored)
	14	host precedence violation	(ignored)
	15	precedence cutoff in effect	(ignored)
4	0	source quench	kernel for TCP, ignored by UDP
5	0	redirect:	
	0	redirect for network	kernel updates routing table
	1	redirect for host	kernel updates routing table
	2	redirect for type-of-service and network	kernel updates routing table
	3	redirect for type-of-service and host	kernel updates routing table
8	0	echo request	kernel generates reply
9	0	router advertisement	user process
10	0	router solicitation	user process
11	0	time exceeded:	
	0	TTL equals 0 during transit	user process
	1	TTL equals 0 during reassembly	user process
12	0	parameter problem:	
	0	IP header bad (catchall error)	"Protocol not available"
	1	required option missing	"Protocol not available"
13	0	timestamp request	kernel generates reply
14	0	timestamp reply	user process
15	0	information request (obsolete)	(ignored)
16	0	information reply (obsolete)	user process
17	0	address mask request	kernel generates reply
18	0	address mask reply	user process

ICMPv4 Message	ICMPv6 Equivalent
Destination Unreachable-Network Unreachable (Type 3, Code 0)	Destination Unreachable-No Route to Destination (Type 1, Code 0)
Destination Unreachable-Host Unreachable (Type 3, Code 1)	Destination Unreachable-Address Unreachable (Type 1, Code 3)
Destination Unreachable-Protocol Unreachable (Type 3, Code 2)	Parameter Problem-Unrecognized Next Header Type Encountered (Type 4, Code 1)
Destination Unreachable-Port Unreachable (Type 3, Code 3)	Destination Unreachable-Port Unreachable (Type 1, Code 4)
Destination Unreachable-Fragmentation Needed and DF Set (Type 3, Code 4) (as specified in RFC 1191)	Packet Too Big (Type 2, Code 0)
Destination Unreachable-Communication with Destination Host Administratively Prohibited (Type 3, Code 10)	Destination Unreachable-Communication with Destination Administratively Prohibited (Type 1, Code 1)
Source Quench (Type 4, Code 0)	This message is not present in IPv6.
Redirect (Type 5, Code 0)	Neighbor Discovery Redirect message (Type 137, Code 0). For more information, see Chapter 6.
Time Exceeded-TTL Exceeded in Transit (Type 11, Code 0)	Time Exceeded-Hop Limit Exceeded in Transit (Type 3, Code 0)
Time Exceeded-Fragment Reassembly Time Exceeded (Type 11, Code 1)	Time Exceeded-Fragment Reassembly Time Exceeded (Type 3, Code 1)
Parameter Problem (Type 12, Code 0)	Parameter Problem (Type 4, Code 0 or Code 2)

Time Exceeded

- When the Time to Live (TTL) field in a packet is decremented to 0, an ICMPv4 Time Exceeded message will be sent to the source host.
- ICMPv6 also sends a Time Exceeded message. Instead of the IPv4 TTL field, ICMPv6 uses the IPv6 Hop Limit field to determine if the packet has expired.

```
Pinging 8.8.8.8 with 32 bytes of data:  
Reply from 192.168.1.1: TTL expired in transit.  
Reply from 192.168.1.1: TTL expired in transit.  
Reply from 192.168.1.1: TTL expired in transit.  
Reply from 192.168.1.1: TTL expired in transit.  
  
Ping statistics for 8.8.8.8:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

Note: Time Exceeded messages are used by the **tracert** tool.

ICMP Messages

ICMPv6 Messages

Messaging between an IPv6 router and an IPv6 device, including dynamic address allocation are as follows:

- Router Solicitation (RS) message
- Router Advertisement (RA) message

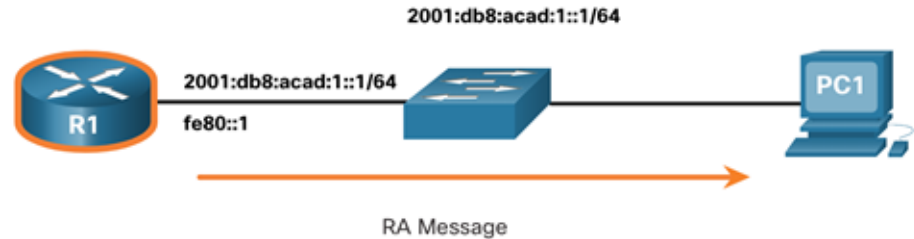
Messaging between IPv6 devices, including duplicate address detection and address resolution are as follows:

- Neighbor Solicitation (NS) message
- Neighbor Advertisement (NA) message

Type	Code	Description
1		destination unreachable
	0	no route to destination
	1	communication with destination administratively prohibited
	2	beyond scope of source address
	3	address unreachable
2	4	port unreachable
	0	packet too big
3		time exceeded
	0	hop limit exceeded in transit
	1	fragment reassembly time exceeded
4		parameter problem
	0	erroneous header field encountered
	1	unrecognized Next Header type encountered
	2	unrecognized IPv6 option encountered
128	0	echo request
129	0	echo reply
130	0	group membership query
131	0	group membership reply
132	0	group membership reduction
133	0	router solicitation
134	0	router advertisement
135	0	neighbor solicitation
136	0	neighbor advertisement
137	0	redirect

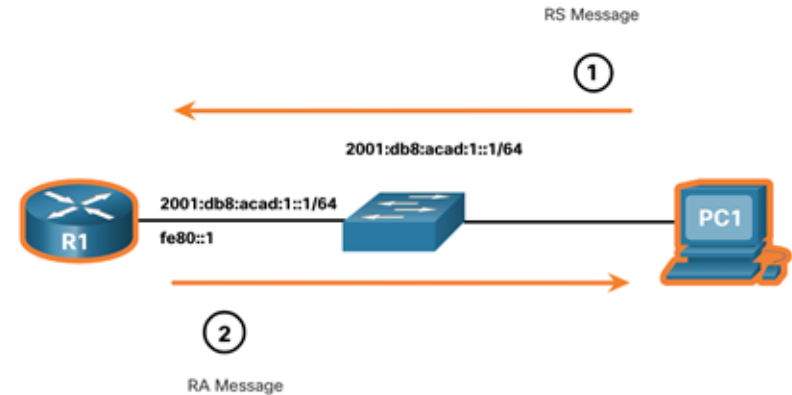
ICMPv6 Messages (Cont.)

- RA messages are sent by IPv6-enabled routers every 200 seconds to provide addressing information to IPv6-enabled hosts.
- RA message can include addressing information for the host such as the prefix, prefix length, DNS address, and domain name.
- A host using Stateless Address Autoconfiguration (SLAAC) will set its default gateway to the link-local address of the router that sent the RA.



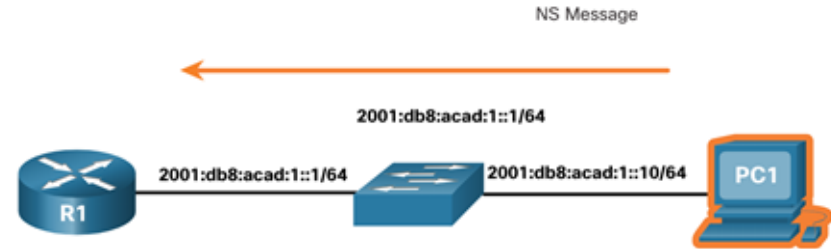
ICMPv6 Messages (Cont.)

- An IPv6-enabled router will also send out an RA message in response to an RS message.
- In the figure, PC1 sends a RS message to determine how to receive its IPv6 address information dynamically.
 - R1 replies to the RS with an RA message.
 - PC1 sends an RS message, "Hi, I just booted up. Is there an IPv6 router on the network? I need to know how to get my IPv6 address information dynamically."
 - R1 replies with an RA message. "Hi all IPv6-enabled devices. I'm R1 and you can use SLAAC to create an IPv6 global unicast address. The prefix is 2001:db8:acad:1::/64. By the way, use my link-local address fe80::1 as your default gateway."



ICMPv6 Messages (Cont.)

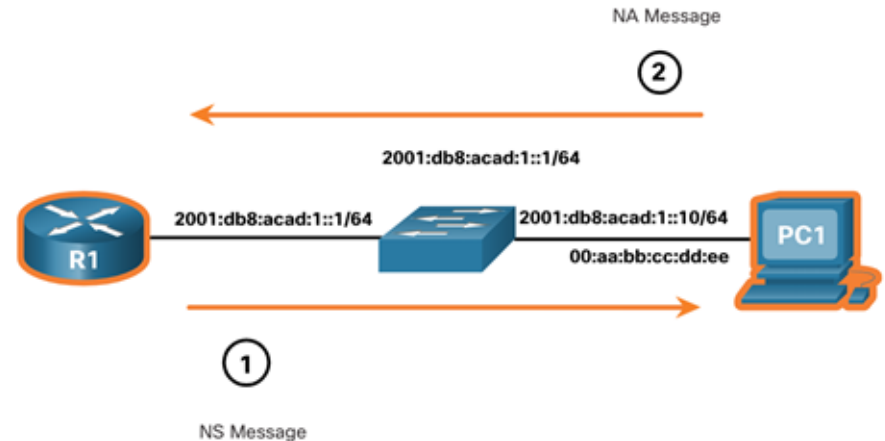
- A device assigned a global IPv6 unicast or link-local unicast address, may perform duplicate address detection (DAD) to ensure that the IPv6 address is unique.
- To check the uniqueness of an address, the device will send an NS message with its own IPv6 address as the targeted IPv6 address.
- If another device on the network has this address, it will respond with an NA message notifying to the sending device that the address is in use.



Note: DAD is not required, but RFC 4861 recommends that DAD is performed on unicast addresses.

ICMPv6 Messages (Cont.)

- To determine the MAC address for the destination, the device will send an NS message to the solicited node address.
- The message will include the known (targeted) IPv6 address. The device that has the targeted IPv6 address will respond with an NA message containing its Ethernet MAC address.
- In the figure, R1 sends a NS message to 2001:db8:acad:1::10 asking for its MAC address.



13.2 Ping and Traceroute Tests

Ping – Test Connectivity

- The **ping** command is an IPv4 and IPv6 testing utility that uses ICMP echo request and echo reply messages to test connectivity between hosts and provides a summary that includes the success rate and average round-trip time to the destination.

```
C:\Users\LCM>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=8ms TTL=115
Reply from 8.8.8.8: bytes=32 time=8ms TTL=115
Reply from 8.8.8.8: bytes=32 time=8ms TTL=115
Reply from 8.8.8.8: bytes=32 time=8ms TTL=115

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 8ms, Average = 8ms

C:\Users\LCM>
```

```
S1#ping 192.168.20.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms
```

```
R1#ping 2001:db8:acad:1::2

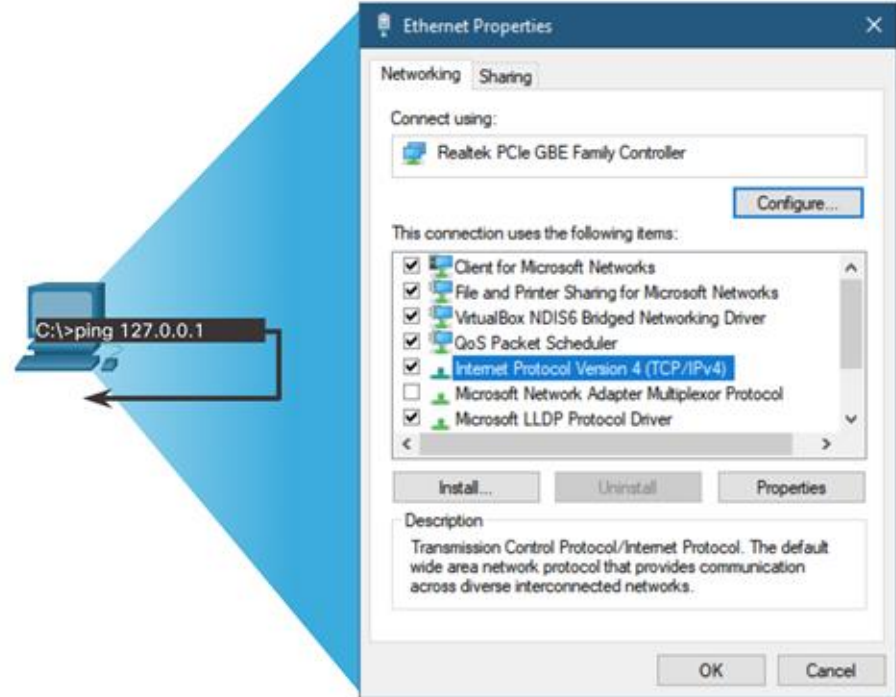
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:db8:acad:1::2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```


Ping and Traceroute Tests

Ping the Loopback

Ping can be used to test the internal configuration of IPv4 or IPv6 on the local host. To do this, **ping** the local loopback address of 127.0.0.1 for IPv4 (:::1 for IPv6).

- A response from 127.0.0.1 for IPv4, or :::1 for IPv6, indicates that IP is properly installed on the host.
- An error message indicates that TCP/IP is not operational on the host.

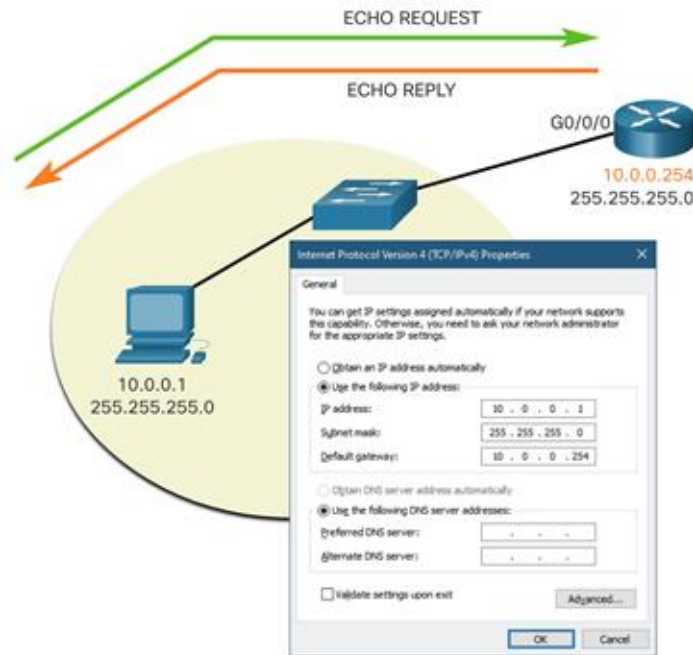


Ping the Default Gateway

The **ping** command can be used to test the ability of a host to communicate on the local network.

The default gateway address is most often used because the router is normally always operational.

- A successful **ping** to the default gateway indicates that the host and the router interface serving as the default gateway are both operational on the local network.
- If the default gateway address does not respond, a **ping** can be sent to the IP address of another host on the local network that is known to be operational.

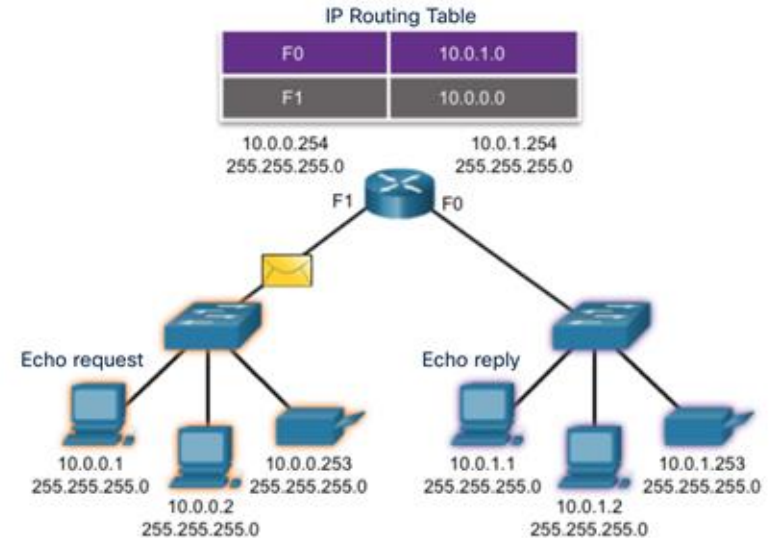


Ping a Remote Host

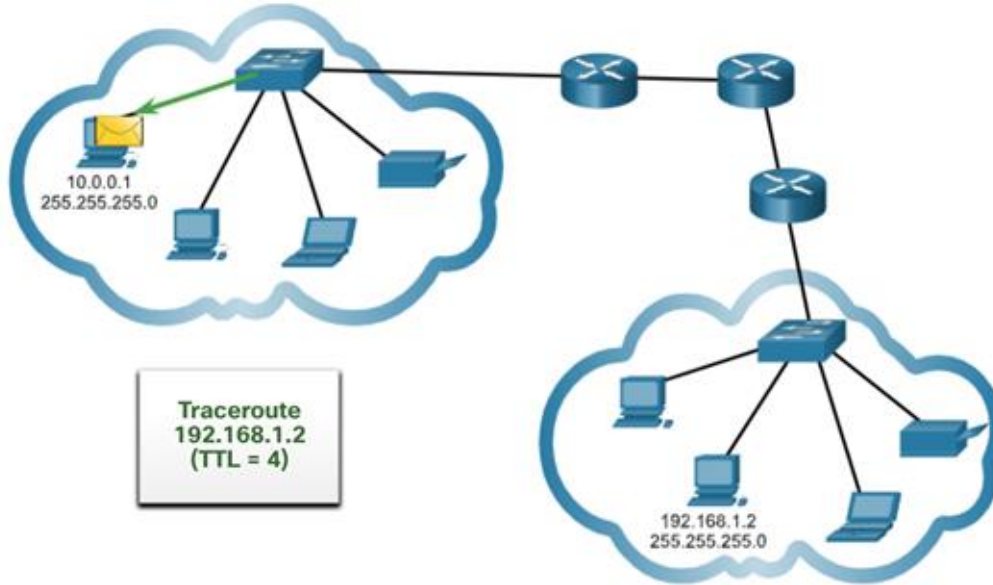
Ping can also be used to test the ability of a local host to communicate across an internetwork.

A local host can ping a host on a remote network. A successful **ping** across the internetwork confirms communication on the local network.

Note: Many network administrators limit or prohibit the entry of ICMP messages therefore, the lack of a **ping** response could be due to security restrictions.



Traceroute – Test the Path



Traceroute

```
C:\Users\LCM>tracert 8.8.8.8
```

```
Tracing route to dns.google [8.8.8.8]  
over a maximum of 30 hops:
```

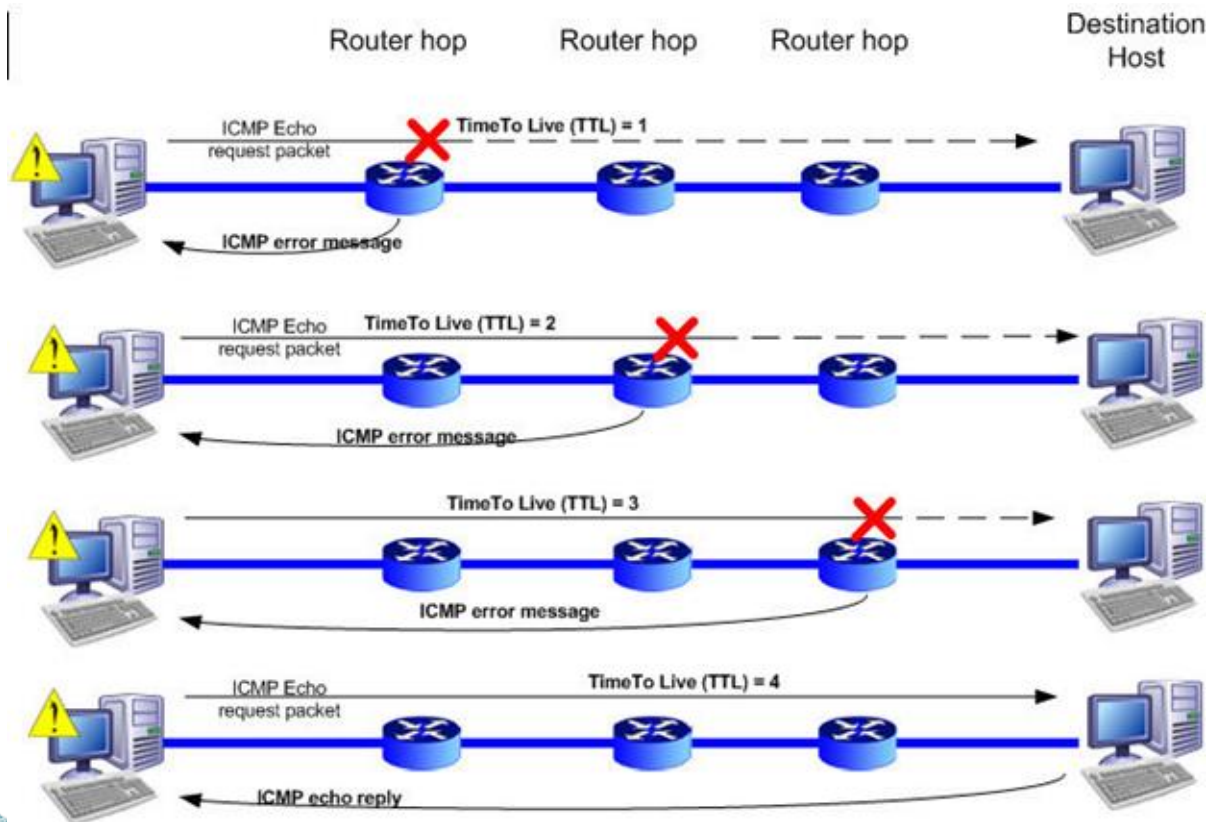
1	<1 ms	<1 ms	<1 ms	10.0.0.1
2	<1 ms	<1 ms	<1 ms	192.168.196.161
3	1 ms	1 ms	<1 ms	192.168.196.41
4	1 ms	1 ms	1 ms	192.168.197.237
5	1 ms	1 ms	1 ms	192.168.193.193
6	1 ms	1 ms	1 ms	xe3100-958.rt.nxd.buh.ro.retn.net [87.245.238.73]
7	9 ms	9 ms	9 ms	et411-4.rt.tlp.sof.bg.retn.net [87.245.232.43]
8	9 ms	8 ms	8 ms	74.125.51.214
9	14 ms	13 ms	12 ms	216.239.62.49
10	8 ms	8 ms	8 ms	142.251.227.249
11	8 ms	8 ms	8 ms	dns.google [8.8.8.8]

```
Trace complete.
```

```
C:\Users\LCM>
```

Ping and Traceroute Tests

Traceroute





Telecom Academy



Module 9: Address Resolution

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Cristian Deacu

Introduction to Networks v7.0
(ITN)



9.1 MAC and IP

Destination on Same Network

There are two primary addresses assigned to a device on an Ethernet LAN:

- **Layer 2 physical address (the MAC address)** – Used for NIC to NIC communications on the same Ethernet network.
- **Layer 3 logical address (the IP address)** – Used to send the packet from the source device to the destination device.

Layer 2 addresses are used to deliver frames from one NIC to another NIC on the same network. If a destination IP address is on the same network, the destination MAC address will be that of the destination device.



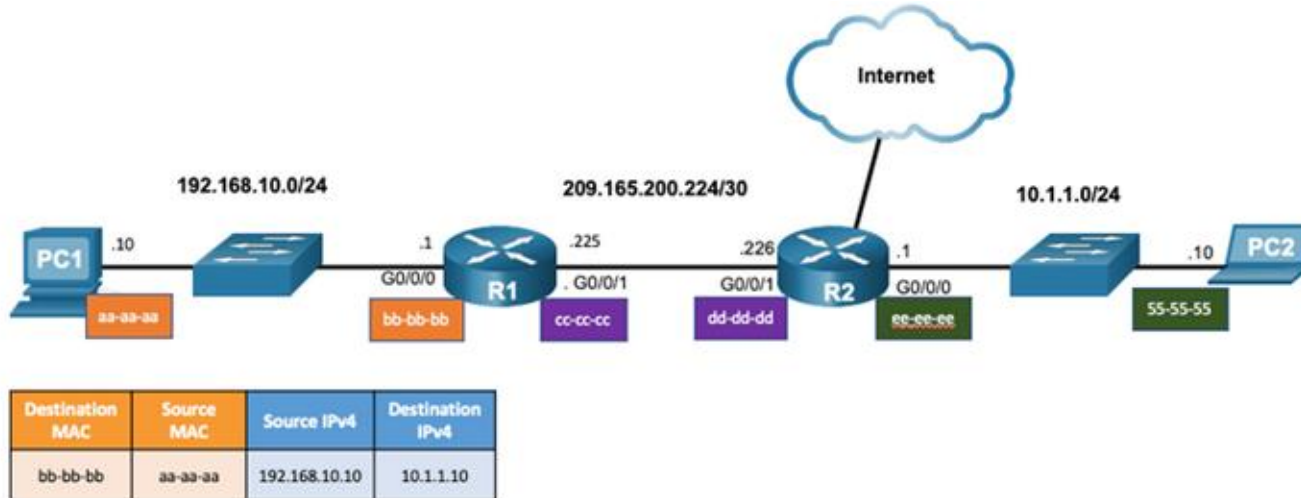
Destination MAC	Source MAC	Source IPv4	Destination IPv4
55-55-55	aa-aa-aa	192.168.10.10	192.168.10.11

MAC and IP

Destination on Remote Network

When the destination IP address is on a remote network, the destination MAC address is that of the default gateway.

- ARP is used by IPv4 to associate the IPv4 address of a device with the MAC address of the device NIC.
- ICMPv6 is used by IPv6 to associate the IPv6 address of a device with the MAC address of the device NIC.



9.2 ARP

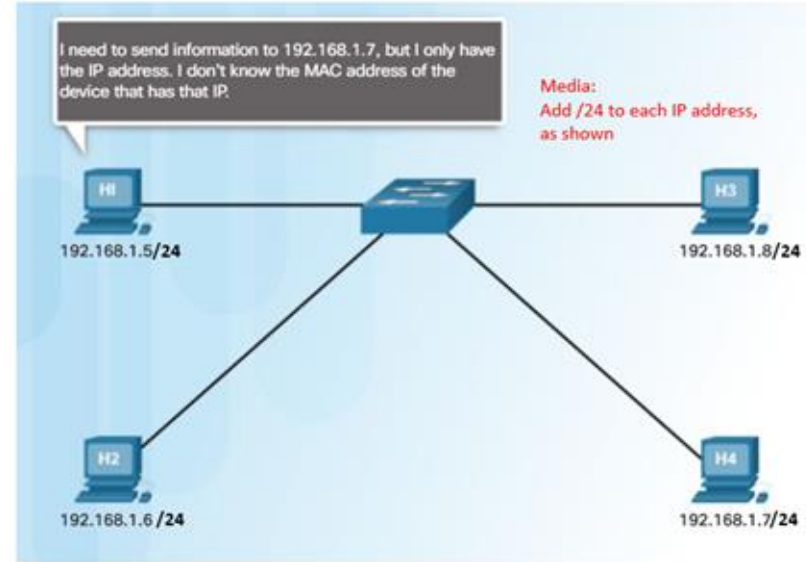
ARP

ARP Overview

A device uses ARP to determine the destination MAC address of a local device when it knows its IPv4 address.

ARP provides two basic functions:

- Resolving IPv4 addresses to MAC addresses
- Maintaining an ARP table of IPv4 to MAC address mappings



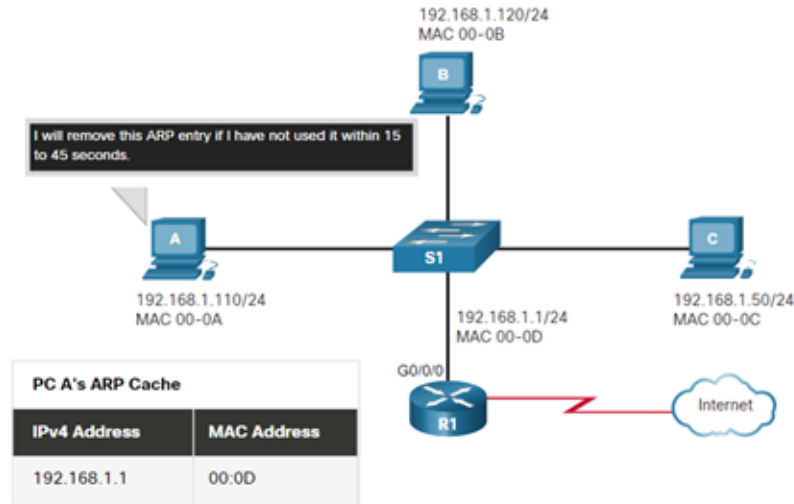
ARP

Removing Entries from an ARP Table

Entries in the ARP table are not permanent and are removed when an ARP cache timer expires after a specified period of time.

The duration of the ARP cache timer differs depending on the operating system.

ARP table entries can also be removed manually by the administrator.



Note: MAC addresses are shortened for demonstration purposes.

ARP

ARP Tables on Networking Devices

The `show ip arp` command displays the ARP table on a Cisco router.

The `arp -a` command displays the ARP table on a Windows 10 PC.

```
R1# show ip arp
```

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	192.168.10.1	-	a0e0.af0d.e140	ARPA	GigabitEthernet0/0/0

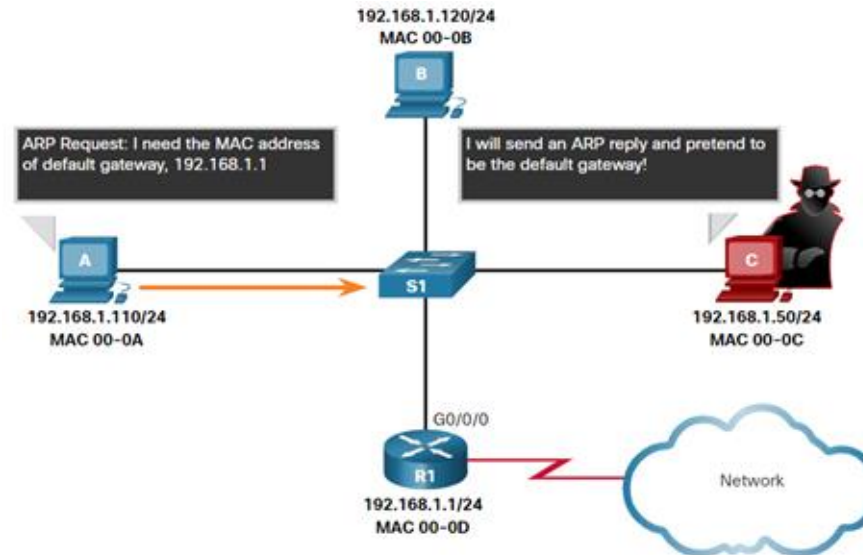
```
C:\Users\PC> arp -a
```

```
Interface: 192.168.1.124 --- 0x10
```

Internet Address	Physical Address	Type
192.168.1.1	c8-d7-19-cc-a0-86	dynamic
192.168.1.101	08-3e-0c-f5-f7-77	dynamic

ARP Issues – ARP Broadcasting and ARP Spoofing

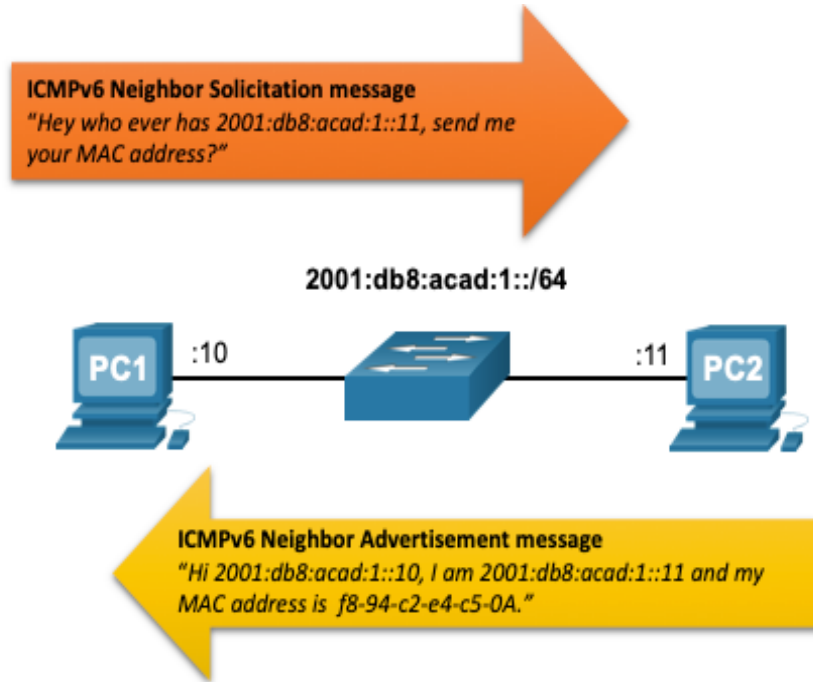
ARP requests are received and processed by every device on the local network.
Excessive ARP broadcasts can cause some reduction in performance.
ARP replies can be spoofed by a threat actor to perform an ARP poisoning attack.
Enterprise level switches include mitigation techniques to protect against ARP attacks.



Note: MAC addresses are shortened for demonstration purposes.

9.3 IPv6 Neighbor Discovery

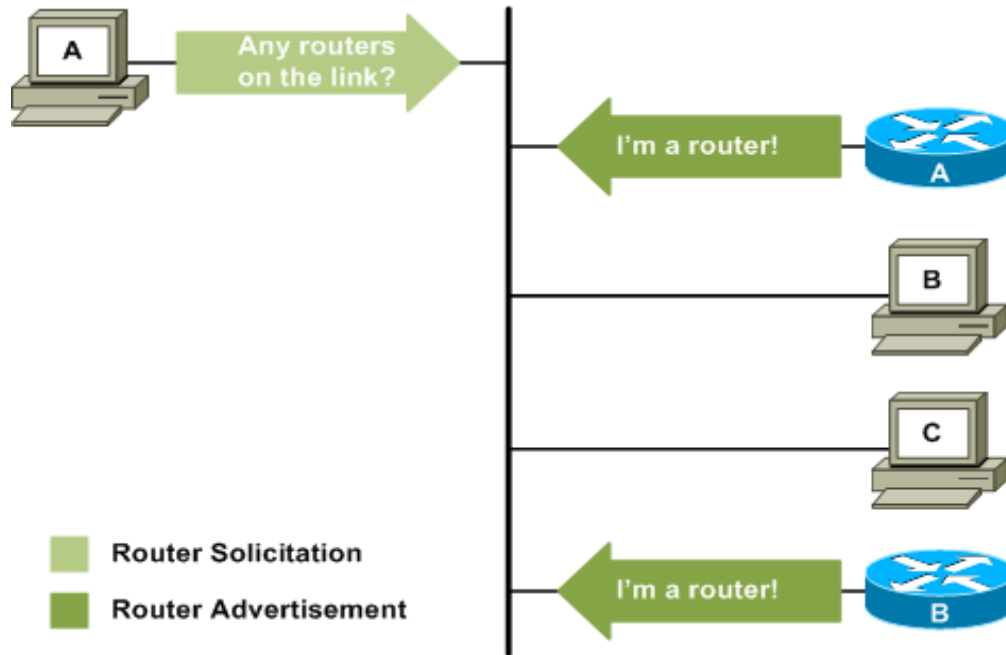
IPv6 Neighbor Discovery – Address Resolution



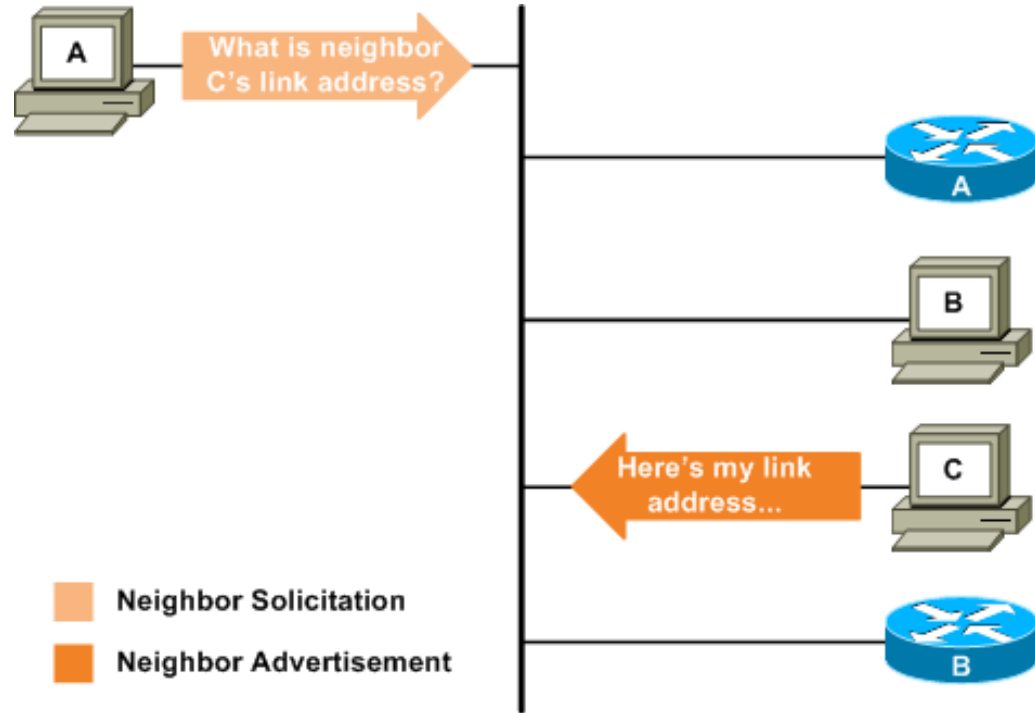
IPv6 devices use ND to resolve the MAC address of a known IPv6 address.

ICMPv6 Neighbor Solicitation messages are sent using special Ethernet and IPv6 multicast addresses.

- **Router Solicitation** (type 133) and **Router Advertisement** (type 134)



- **Neighbor Solicitation** (type 135) and **Neighbor Advertisement** (type 136)



No.	Time	Source	Destination	Protocol	Length	Info
5	14.8378490	2001:db2::7729:c0ad	ff02::1:ff5c:7a92	ICMPv6	86	Neighbor Solicitation for 2001:db2::1f5c:7a92 from c0:02:21:68:00:00
[-] Frame 5: 86 bytes on wire (688 bits), 86 bytes captured (688 bits) on interface 0						
[-] Ethernet II, Src: c0:02:21:68:00:00 (c0:02:21:68:00:00), Dst: IPv6mcast_ff5c:7a:92 (33:33:ff:5c:7a:92)						
[-] Internet Protocol Version 6, Src: 2001:db2::7729:c0ad (2001:db2::7729:c0ad), Dst: ff02::1:ff5c:7a92 (ff02::1:ff5c:7a92)						
[-] 0110 = Version: 6						
[-] 1110 0000 = Traffic class: 0x000000e0						
.... 0000 0000 0000 0000 0000 = Flowlabel: 0x00000000						
Payload length: 32						
Next header: ICMPv6 (58)						
Hop limit: 255						
Source: 2001:db2::7729:c0ad (2001:db2::7729:c0ad)						
Destination: ff02::1:ff5c:7a92 (ff02::1:ff5c:7a92)						
[Source GeoIP: Unknown]						
[Destination GeoIP: unknown]						
[-] Internet Control Message Protocol v6						
Type: Neighbor Solicitation (135)						
Code: 0						
Checksum: 0xf019 [correct]						
Reserved: 00000000						
Target Address: 2001:db2::1f5c:7a92 (2001:db2::1f5c:7a92)						
[-] ICMPv6 option (Source link-layer address : c0:02:21:68:00:00)						

